

MARKETING FUNNEL & CONVERSION PERFORMANCE ANALYSIS

Data Science & Analytics Internship

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- Microsoft SQL Server
 - Power BI
- Microsoft Excel

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EXECUTIVE SUMMARY

The **Marketing Funnel & Conversion Performance Analysis** project focuses on evaluating customer behavior throughout the online purchasing journey and identifying opportunities to improve conversion performance. The analysis examines how customers progress through different stages of the marketing funnel, including **Browse, Add to Cart, Checkout, and Purchase**, to understand customer engagement and purchasing patterns.

This project was developed using **Microsoft SQL Server** for data analysis, **Power BI** for interactive dashboard visualization, and **Microsoft Excel** for data preparation. The dataset consists of **10,000 customer sessions** containing information such as user interactions, marketing channels, devices, regions, product categories, and revenue. SQL queries were used to calculate key performance indicators and generate meaningful business insights, while Power BI was used to visualize the results through an interactive dashboard.

The analysis revealed a **total revenue of ₹277,323.1** and an overall **conversion rate of 10.04%**, indicating that only a small percentage of customer sessions resulted in successful purchases. The funnel analysis highlighted significant customer drop-offs between each stage of the purchasing journey. Additionally, **Google Ads** emerged as the highest-performing marketing channel, the **South** region generated the highest revenue, and the **Electronics** category recorded the highest customer activity.

The insights generated from this project can help organizations optimize marketing strategies, improve customer experience, increase conversion rates, and support data-driven decision-making for sustainable business growth.

KPI SUMMARY

The following Key Performance Indicators (KPIs) provide a concise overview of the marketing funnel's overall performance. These metrics summarize customer engagement, conversion efficiency, and revenue generation, serving as the foundation for the detailed analysis presented in the subsequent sections.

KPI	Value
Total Customer Sessions	10,000
Total Purchases	1,004
Conversion Rate	10.04%
Total Revenue	₹277,323.1
Purchase Revenue	₹277,323.1
Browse Events	10,000
Add to Cart Events	6,949
Checkout Events	3,456
Purchase Events	1,004
Top Marketing Channel	Google Ads
Highest Revenue Region	South
Most Active Device	Desktop
Top Product Category	Electronics

INTRODUCTION

Marketing Funnel & Conversion Performance Analysis is a business analytics process used to evaluate how customers progress through different stages of the purchasing journey, from initial product discovery to the final purchase. By analyzing customer interactions at each stage, organizations can identify where potential customers leave the funnel and implement strategies to improve conversion rates and overall business performance.

In today's digital marketplace, businesses invest in multiple marketing channels such as Google Ads, Email Marketing, Organic Search, and Social Media to attract customers. However, not every visitor completes a purchase. Many customers browse products, add items to their carts, or proceed to checkout without completing the transaction. Understanding these customer behaviors is essential for improving marketing effectiveness, enhancing user experience, and maximizing revenue.

This project uses **Microsoft SQL Server** for data analysis and **Power BI** for interactive dashboard visualization. The dataset contains **10,000 customer sessions** with information related to user interactions, marketing channels, devices, regions, product categories, and revenue. Through SQL queries and data visualization, the project evaluates customer engagement, conversion performance, revenue generation, and marketing effectiveness.

Objectives of the Project

- Analyze customer interactions across the marketing funnel.
- Measure customer conversion from Browse to Purchase.
- Evaluate total revenue and purchasing performance.
- Identify customer drop-off points within the funnel.

- Compare the performance of different marketing channels, regions, devices, and product categories.
 - Develop an interactive Power BI dashboard for business reporting.
 - Generate actionable insights to improve customer conversion and support data-driven decision-making.
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DATASET OVERVIEW

The dataset used in this project contains customer interaction data collected throughout the online purchasing journey. It is designed to analyze customer behavior, evaluate marketing funnel performance, and identify opportunities to improve conversion rates and revenue generation. The dataset consists of **10,000 customer sessions**, with each record representing a customer event during an online shopping session.

The dataset includes the following attributes:

- **User ID** – Unique identifier for each customer.
- **Session ID** – Unique identifier for each customer session.
- **Event Time** – Time at which a customer interaction occurred.
- **Event** – Customer activity, such as Browse, Add to Cart, Checkout, or Purchase.
- **Device** – Device used by the customer (Desktop, Mobile, or Tablet).
- **Region** – Customer's geographical location.
- **Channel** – Marketing source, including Google Ads, Email, Organic Search, and Social Media.
- **Product Category** – Category of the product viewed or purchased.
- **Revenue** – Revenue generated from completed purchases.
- **Bonus Flag** – Indicates whether a promotional offer was applied.

The dataset was imported into **Microsoft SQL Server** for analysis and **Power BI** for interactive dashboard visualization. The insights obtained from this dataset help evaluate

customer engagement, marketing performance, and conversion efficiency, enabling data-driven business decision-making.

SESSION ANALYSIS

Session analysis provides an overview of customer activity recorded during the analysis period. The dataset contains **10,000 unique customer sessions**, representing individual visits to the online platform. Each session captures one or more customer interactions, including browsing products, adding items to the cart, proceeding to checkout, or completing a purchase.

Analyzing customer sessions helps organizations understand website traffic, measure user engagement, and evaluate the effectiveness of marketing campaigns in attracting visitors. A higher number of sessions indicates strong customer interest and provides a larger opportunity for converting visitors into paying customers.

The session analysis serves as the foundation for the marketing funnel, as every customer journey begins with a session. By examining customer activity across these sessions, businesses can identify engagement patterns, evaluate customer behavior, and develop strategies to improve conversion rates and overall business performance.

FUNNEL ANALYSIS

Marketing funnel analysis evaluates how customers progress through different stages of the purchasing journey, from their initial interaction to the final purchase. The dataset recorded **10,000 Browse events**, of which **6,949 customers** added products to their cart, **3,456 customers** proceeded to checkout, and **1,004 customers** successfully completed a purchase.

The analysis revealed an overall **conversion rate of 10.04%**, indicating that approximately one out of every ten customer sessions resulted in a successful transaction. A significant reduction in customer activity was observed between each stage of the funnel, highlighting areas where potential customers abandoned the purchasing process.

Understanding these drop-off points enables businesses to identify weaknesses in the customer journey and implement strategies to improve user experience, simplify the checkout process, and increase conversion rates. Funnel analysis plays a crucial role in evaluating marketing effectiveness and supporting data-driven business decisions aimed at maximizing customer retention and revenue generation.

REVENUE ANALYSIS

Revenue analysis evaluates the financial performance of the marketing funnel by measuring the income generated from completed customer purchases. The analysis revealed a **total revenue of ₹277,323.1**, indicating the overall value generated through successful customer transactions during the analysis period.

The revenue generated reflects the effectiveness of the marketing campaigns in converting customer interest into actual sales. While the dataset recorded **10,000 customer sessions**, only **1,004** resulted in completed purchases, producing an overall **conversion rate of 10.04%**. This highlights the relationship between customer conversions and revenue generation.

Analyzing revenue helps organizations identify profitable marketing strategies, evaluate business performance, and make informed decisions regarding customer acquisition and retention. By improving customer conversion rates and reducing funnel drop-offs, businesses can increase revenue and achieve sustainable growth.

CHANNEL ANALYSIS

Channel analysis evaluates customer traffic and revenue generated through different marketing channels, including **Google Ads, Email, Organic Search, and Social Media**. Understanding channel performance helps organizations identify which marketing sources are most effective in attracting customers and generating business revenue.

The analysis revealed that **Google Ads** was the highest-performing marketing channel, generating the greatest revenue among all available channels. The remaining channels, including Email, Organic Search, and Social Media, also contributed to customer acquisition and sales, demonstrating the importance of maintaining a diversified marketing strategy.

Evaluating marketing channel performance enables businesses to optimize advertising budgets, improve campaign effectiveness, and focus on channels that deliver higher customer engagement and conversions. These insights support data-driven marketing decisions and help maximize return on investment (ROI).

REGIONAL ANALYSIS

Regional analysis examines customer activity and revenue generated across different geographical regions, including **North, South, East, and West**. Understanding regional performance enables organizations to identify high-performing markets and develop location-specific marketing strategies.

The analysis showed that the **South** region generated the highest revenue, indicating stronger customer engagement and purchasing activity compared to the other regions. The North, East, and West regions also contributed significantly to overall business performance, reflecting a balanced customer distribution across multiple locations.

Evaluating regional performance helps businesses optimize regional marketing campaigns, allocate resources effectively, and identify areas with growth

potential. These insights support strategic decision-making by enabling organizations to focus on regions that offer the greatest opportunities for customer acquisition and revenue generation.

DEVICE ANALYSIS

Device analysis evaluates customer interactions across different devices, including **Desktop, Mobile, and Tablet**. Understanding device usage helps organizations identify customer preferences and optimize the user experience across multiple platforms.

The analysis revealed that **Desktop** devices recorded the highest customer activity and generated the greatest revenue, followed by **Tablet** and **Mobile** devices. This indicates that a significant proportion of customers preferred using desktop devices for browsing products and completing purchases.

Analyzing device performance enables businesses to improve website responsiveness, enhance user experience across different platforms, and optimize digital marketing strategies. These insights help organizations ensure a seamless shopping experience, improve customer satisfaction, and increase conversion rates.

PRODUCT CATEGORY ANALYSIS

Product Category Analysis evaluates customer interactions and purchasing behavior across different product categories available on the platform.

Understanding category performance helps organizations identify customer preferences, measure product demand, and optimize inventory and marketing strategies.

The analysis revealed that the **Electronics** category recorded the highest level of customer engagement, followed by other categories such as **Beauty, Fashion,**

Home, and Sports. This indicates that customers showed greater interest in electronics products, making it one of the most significant contributors to overall customer activity and business performance.

Analyzing product category performance enables businesses to identify high-demand products, plan targeted promotional campaigns, and allocate resources more effectively. These insights support data-driven decision-making, improve inventory management, and help maximize revenue by focusing on customer preferences and market demand.

BUSINESS IMPACT

The findings from this Marketing Funnel & Conversion Performance Analysis provide valuable insights that support strategic business decision-making. By analyzing customer behavior across different stages of the marketing funnel, organizations can identify where potential customers leave the purchasing process and implement targeted strategies to improve conversion rates.

The analysis also highlights the performance of marketing channels, regions, devices, and product categories, enabling businesses to optimize marketing campaigns, allocate resources more effectively, and focus on high-performing segments. These insights help improve customer engagement, enhance the overall shopping experience, and maximize the return on marketing investments.

By leveraging data-driven insights, organizations can reduce customer drop-offs, increase revenue, strengthen customer retention, and make informed decisions that contribute to sustainable business growth and long-term success.

CONCLUSION

The **Marketing Funnel & Conversion Performance Analysis** project successfully evaluated customer behavior throughout the online purchasing journey and provided valuable insights into marketing performance, customer engagement, and revenue generation. By analyzing customer interactions across the Browse, Add to Cart, Checkout, and Purchase stages, the project identified key conversion trends and highlighted opportunities to reduce customer drop-offs.

The analysis revealed a total of **10,000 customer sessions**, **1,004 successful purchases**, an overall **conversion rate of 10.04%**, and a **total revenue of ₹277,323.1**. Furthermore, **Google Ads** emerged as the highest-performing marketing channel, the **South** region generated the highest revenue, and the **Electronics** category recorded the greatest customer engagement.

The interactive Power BI dashboard and SQL analysis transformed raw data into meaningful business insights, enabling effective visualization of customer behavior and marketing performance. Overall, this project demonstrates how data analytics and business intelligence can support data-driven decision-making, improve marketing strategies, enhance customer experience, and contribute to sustainable business growth.

DASHBOARD VISUALIZATION

Figure 1: Marketing Funnel & Conversion Performance Dashboard

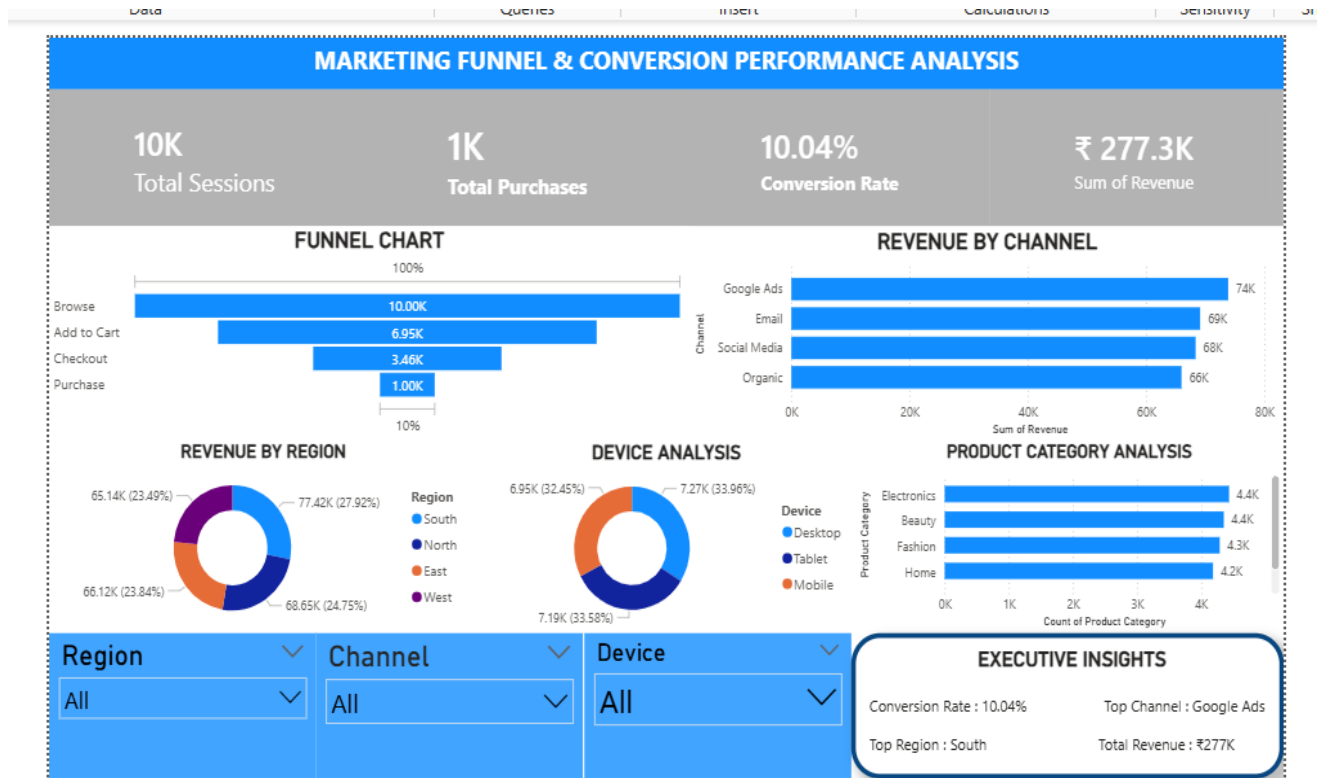


Figure illustrates the interactive Power BI dashboard developed for the Marketing Funnel & Conversion Performance Analysis project. The dashboard presents key performance indicators, customer funnel stages, revenue analysis, channel performance, regional distribution, device usage, and product category insights. It enables decision-makers to monitor marketing effectiveness, identify conversion drop-offs, and make informed, data-driven business decisions.